

Personal Academic Website — Content Sheet

1. Identity & Header

Full name: Pushty Shrimankar

One-line introduction (pick one):

- *Early-universe cosmologist working on gravitational-wave signatures of primordial magnetic fields.*
- *Theoretical cosmologist — inflationary magnetogenesis, stochastic gravitational-wave backgrounds, and observational tests of cosmic isotropy.*
- *Theoretical physicist bridging inflationary modelling and observational cosmology.*

*(Claude info) The first is the cleanest for a site header; the second is better for a PhD-application page because it names all three of your research threads.

Current status: Final-year M.Sc. Physics student at VIT Vellore (2024–2026), completing a master's thesis at the Indian Institute of Astrophysics (IIA), Bengaluru. Applying for PhD positions in theoretical cosmology and gravitational-wave physics for the 2026 admissions cycle.

Professional photo.....

Location: Rajkot, Gujarat, India (currently based in Bengaluru for thesis work at IIA)

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Phone: +91-8200174674

Links:

Profile	Status
LinkedIn	linkedin.com/in/pushty-shrimankar-652198280 — active
GitHub	
ORCID	
Google Scholar	
INSPIRE-HEP	
arXiv author page	https://doi.org/10.48550/arXiv.2604.04436

2. Education

M.Sc. Physics — Vellore Institute of Technology, Vellore · 2024–2026

Current CGPA 8.94 / 10

Coursework: General Relativity, Astrophysics, Statistical Mechanics, Quantum mechanics, Electrodynamics, Classical Mechanics, Mathematical Physics, Atomic and Molecular physics, solid state physics, Plasma physics, Semiconductor devices, nanophysics, electronics, general physics lab

B.Sc. Physics — H. and H. B. Kotak Institute of Science, Rajkot (Saurashtra University) · 2019–2022

CGPA 9/10

Ranked 9th in the university

Final-year project: Solar differential rotation via sunspot tracking (Prof. Hari Om Vats)

3. Specialization

Primary: Early-universe cosmology — inflationary magnetogenesis and the stochastic gravitational-wave background (SGWB) it sources.

Secondary: Observational cosmology — model-independent, data-driven tests of the cosmological principle using Type Ia supernovae and BAO.

Peripheral interests: Modified gravity; fast radio bursts and the CMB; plasma astrophysics.

4. Master's Thesis

Title: *Computational Exploration of Gravitational Wave Imprints from Magnetogenesis During Inflation*

Indian Institute of Astrophysics (IIA), Bengaluru — Visiting Student Programme 2026 · Submitted May 2026

Summary:

Magnetic fields are found everywhere in the universe, from galaxies to the near-empty space between them, and nobody is certain where the first seed fields came from. One possibility is that they were created during inflation, the burst of expansion in the universe's first instant. This thesis works out how strong such primordial fields would be, and whether the gravitational waves they generate could ever be detected. The answer, for the simplest non-helical case, is no, the signal sits about six orders of magnitude below what LISA, DECIGO, or the Einstein Telescope could see, which points toward helical models as the more promising place to look.

Key numbers for the site: $\Omega_{\text{GW}} \approx 4.4 \times 10^{-19}$ for $n = 2$; backreaction constraint $\Omega_{\text{EM}} \approx 2.1 \times 10^{-5} \ll 1$ satisfied.

5. Awards, Scholarships, Exam Ranks

- Ranked **9th** in B.Sc. Physics, Saurashtra University
 - Selected for the **Visiting Student Programme 2026**, Indian Institute of Astrophysics
 - Selected for the **ICTS Summer School in Gravitational Waves**, 2025
 - Selected for the **IUCAA Summer School in Astronomy and Astrophysics**, 2021
 - **[NEED FROM YOU]** GATE / JEST / CSIR-NET / GRE scores and ranks. If you've cleared any of these, it belongs near the top of a PhD-application site — Indian institutes screen on it directly. If you haven't taken them yet, say so and I'll leave the section out rather than have an empty heading.
 - **[NEED FROM YOU]** Any scholarships, fee waivers, or merit stipends (INSPIRE-SHE, KVPY, state merit, VIT scholarship)?
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6. Research Experience

Research Intern (Master's Thesis) — Indian Institute of Astrophysics, Bengaluru · 1st January 2026 to 31st July 2026

Supervisor: Dr. Debika Chowdhury (DST INSPIRE Faculty Fellow, IIA)

Internal supervisor: Dr. Gopal Kashyap (School of Advanced Sciences, VIT)

Modelled the SGWB sourced by non-helical primordial magnetic fields; derived analytic gauge-field solutions across the pre-inflationary, inflationary, and reheating eras; computed $\Omega_{\text{GW}}(f)$ against LISA, ET, DECIGO, BBO, and aLIGO sensitivity curves; carried out the backreaction consistency analysis.

Cosmic Distance Duality Relation and BAO Analysis — VIT Vellore · 2025–2026

Supervisor: Dr. Gopal Kashyap

Lead author on arXiv:2604.04436. Built a model-independent Python pipeline testing directional anisotropy in the CDDR using Pantheon+, BAO (6dFGS, SDSS, BOSS, eBOSS, DES, DESI DR1/DR2), and cosmic chronometers.

Similarity Parameters in Plasma and Gravitational Waves — VIT Vellore · 2025

Supervisor: Dr. Vishwa Bandhu Pathak

Explored self-similar scaling analogies between plasma wave propagation and gravitational-wave signatures.

Stellar and Solar Observations — Neogen Solar and Stellar Observatory, Rajkot · 2024–present

Telescope operation, Messier object tracking, solar spectrographic observation.

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7. Publications & Preprints

Directional Tests of the Cosmic Distance Duality Relation using Pantheon+ and BAO

Pushty Shrimankar, Gopal Kashyap

Published in arXiv:2604.04436 [astro-ph.CO] · Prepared for submission to *JCAP*

<https://arxiv.org/abs/2604.04436>

One-line description for the site: A model-independent search for directional anisotropy in the cosmic distance duality relation, yielding a 95% upper bound of $A < 0.025$ on the dipole amplitude — consistent with isotropy, and stable across supernova calibration choices.

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8. Talks, Posters & Programmes

Upcoming — ICTS programme on primordial magnetic fields

“Magnetic fields in the primordial universe”

Selected participant; presenting a talk on the gravitational-wave signatures of inflationary magnetogenesis

DATE & TIME

30 November 2026 to 04 December 2026

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Schools & workshops attended:

- ICTS Summer School in Gravitational Waves, 2025 — detection pipelines, waveform modelling, mock-signal analysis in Python
- IUCAA Summer School in Astronomy and Astrophysics, 2021 — gravitational lensing, GW fundamentals, cosmological time dilation
- Photonics Lecture Series, IIT Guwahati, 2021 (online)
- Poster presentation — *Computational Exploration of Gravitational Wave Imprints from Magnetogenesis During Inflation*, VIT

9. Featured Projects

Project 1 — Is the distance duality relation the same in every direction?

The question. The Etherington reciprocity relation, $\eta(z) = D_L / [(1+z)^2 D_A] = 1$, follows from pure geometry and holds in any metric theory of gravity. But nearly every test of it

assumes the universe looks the same in all directions. If it doesn't — and radio-source dipoles and supernova data both hint that it might not — those tests could be quietly biased. Does the relation itself depend on where you look?

My approach. I combined 960 Pantheon+ supernovae at $z > 0.1$ with twelve BAO measurements spanning 6dFGS through DESI DR2, plus 32 cosmic chronometer $H(z)$ points. Rather than assuming Λ CDM, I reconstructed the BAO-derived luminosity distance with Gaussian Process Regression — a Matérn $\nu = 3/2$ kernel for $H(z)$, an RBF kernel for D_L^{BAO} , both chosen by 5-fold cross-validation. I then extended the standard parameterization to $\eta(z, \hat{n}) = (1 + \eta_1 z)(1 + \hat{n} \cdot D)$, propagated the full Pantheon+ covariance into luminosity-distance space, and sampled the posterior with emcee under three different supernova calibrations (free M_B , DB23, SH0ES).

What I found. No anisotropy — with a caveat that turned out to be the more interesting result. The dipole amplitude is constrained to $A < 0.025$ at 95%, stable across all three calibrations. But the null tests showed why that stability matters: shuffling supernova magnitudes while keeping the real sky positions produces a *spurious* dipole peaking at $A \approx 0.048$, six times the observed value, purely from uneven survey coverage. Replacing the positions with isotropic random vectors drops that to $A \approx 0.017$. In other words, survey geometry alone can manufacture an anisotropy signal larger than anything in the data, which is a caution for any dipole search that skips this step.

Figure to use. Figure 6 — the null-test histogram, with the green (real sky geometry) and blue (isotropic) distributions and the observed $A = 0.008$ marked as a vertical line. It tells the whole story in one image. *Alternative:* Figure 1(a), the Mollweide sky projection, if you want something more visually striking for a homepage thumbnail.

Project 2 — Can gravitational waves from primordial magnetic fields ever be detected?

The question. Electromagnetism is conformally invariant in an FLRW universe, so electromagnetic fields can't be amplified during inflation without breaking that invariance. The standard fix is a time-dependent coupling $J(\varphi) \propto a^{-n}$ between the inflaton and the gauge field. Those amplified fields carry anisotropic stress, which sources gravitational waves. Is the resulting background strong enough for LISA, DECIGO, or the Einstein Telescope to see?

My approach. I derived the gauge-field mode equation in Coulomb gauge and solved it analytically across three eras — the pre-inflationary Bunch–Davies vacuum, inflation (Hankel index $\nu = n + 1/2$), and reheating ($\mu = 2n + 1/2$) — matching the mode function and its derivative at the end of inflation. From these I computed the electric and magnetic power spectra $P_E(k)$ and $P_B(k)$, evaluated them numerically in Python using the exact half-integer Hankel expansions, and checked the backreaction condition $\Omega_{\text{EM}} \ll 1$ before converting to a gravitational-wave energy density.

What I found. For $n = 2$ the magnetic spectrum is nearly scale-invariant — the observationally favoured case, since it can seed galactic dynamos — and $\Omega_{\text{EM}} \approx 2.1 \times 10^{-5}$,

comfortably satisfying backreaction. But the payoff is a gravitational-wave background of only $\Omega_{\text{GW}} \approx 4.4 \times 10^{-19}$, roughly six orders of magnitude below every current and planned detector. The mirror case $n = -2$ gives a scale-invariant *electric* spectrum whose energy density blows past unity at small scales, violating consistency outright. So the non-helical mechanism is self-consistent and invisible at the same time — which is precisely the argument for turning to helical models, where inverse cascade pushes Ω_{GW} up toward 10^{-13} .

Figure to use. Figure 3.4 — Ω_{GW} plotted against the LISA, DECIGO, BBO, ET, and aLIGO sensitivity curves. The gap is the point. *Supporting figure:* Figure 3.1, the P_B / P_E spectra, showing the flat magnetic spectrum against the falling electric one.

Project 3 — Scaling analogies between plasma waves and gravitational waves

The question. Do the self-similar scaling laws that govern wave propagation in plasmas have structural analogues in gravitational-wave propagation, and can similarity parameters transfer between the two?

The approach and findings are thinner in what you shared than the other two projects.

► **[NEED FROM YOU]** A few sentences on what you actually did and concluded, plus any plot. If it stayed exploratory and produced no figure, my recommendation is to drop it from the featured projects and keep it as a one-line entry under Research Experience instead. Three strong projects beat two strong and one vague — a reviewer who clicks into a thin third project may discount the first two.

10. Technical Skills

Languages: Python, Mathematica

Scientific stack: NumPy, SciPy, Astropy, scikit-learn, Matplotlib

Cosmology & GW tools: emcee, Cobaya, LALSuite, HEALPix / Mollweide projection

Methods: Gaussian Process Regression, MCMC and Bayesian inference, Gelman–Rubin convergence diagnostics, full-covariance likelihood analysis, null-hypothesis testing with survey-preserving mocks, cosmological perturbation theory, WKB and Hankel-function mode analysis

► **[NEED FROM YOU]** LaTeX (assume yes — list it), Git/version control, HPC or cluster experience at IIA, C/C++ or Fortran, any of CAMB/CLASS/GetDist/ChainConsumer/corner. Also worth adding a line if you’ve used `solve_ivp` for stiff ODE integration, since that’s a transferable skill outside astro.

11. Observational & Lab Experience

- 8–12 inch reflector telescopes — operation, alignment, Messier object tracking (Neogen Observatory, Rajkot)
- Solar spectrographic observations
- Sunspot tracking and photospheric rotation measurement (B.Sc. project)
- Annual participant, Messier Marathon — Big Bang Astronomy Club, Rajkot

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12. Theory Areas

- General Relativity and FLRW cosmology
- Inflationary cosmology, slow-roll formalism, and reheating
- Inflationary magnetogenesis, conformal invariance breaking, and gauge-field dynamics in curved spacetime
- Stochastic gravitational-wave backgrounds and detector sensitivity curves
- Gravitational waves, supernovae and other transients
- Black hole mergers, kilonova, love numbers
- Cosmological distance measures, the Etherington relation, and the cosmological principle
- Bayesian statistics, MCMC, and Gaussian-process reconstruction in a cosmological context

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13. Documents

14. Writing, Outreach & Teaching

- Science outreach volunteer, IIA Open Day — explaining astrophysics to school students and the general public

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